Web Programming Midterm Project

My Digital Guide: Interactive Website

Deadline: 2026/11/16 11:59pm

1. Project Overview

Create a functional, visually appealing, and interactive website on a topic of your choice. Your website should provide useful information or services to its visitors. You may choose any topic that interests you.

Example Topics:

- Travel Guide
- Restaurant or Food Guide
- University Survival Guide
- Personal Fitness Guide
- Pet Care Guide
- Music or Movie Recommendation Website
- Coffee Shop or Small Business Website
- Event or Activity Guide

You are encouraged to be **creative** and **develop your own ideas**.

2. Technical Requirements

Your website must be developed **using HTML, CSS, and JavaScript only**.

The following requirements must be satisfied:

A. HTML — Structure and Content

Your website must include:

1. Appropriate headings and paragraphs.
2. At least four interconnected HTML pages.
3. A navigation menu on every page that allows users to navigate directly to all other pages.

4. Create at least one 2×2 grid layout (2 rows and 2 columns) using HTML ** ** elements and CSS. Each cell must contain meaningful content.
5. Images with appropriate alternative text (`alt`).
6. Ordered or unordered lists.
7. At least one meaningful HTML table.
8. Internal and external hyperlinks.
9. Proper HTML document structure.

Your content should be well-organized, meaningful, and relevant to your chosen topic.

B. CSS — Design and Layout

Your website must:

1. Use an external CSS file.
2. Have a consistent visual design across all pages (e.g., consistent color schemes, fonts, navigation menus, headers, and footers).
3. Apply appropriate colors, fonts, spacing, and backgrounds.
4. Use CSS to organize the layout of the website.
5. Include hover effects for interactive elements.
6. Maintain good readability and usability.

You are encouraged to explore additional CSS properties to improve your design.

C. JavaScript — Interactivity

Your website must include at least two meaningful interactive features implemented using JavaScript.

Examples include:

- Interactive buttons that change webpage content.

- A simple calculator.

- A shopping cart that calculates the total price.

- An interactive quiz.

- A dynamic table generated using JavaScript.

The two interactive features must serve different purposes and demonstrate the use of JavaScript and DOM manipulation.

3. Code Explanation

In addition to your website, you must submit a **Code Explanation Report** (PDF).

Your report must include:

A. Project Introduction

- What is the purpose of your website?
- Who are the intended users?
- Why did you choose this topic?

B. Website Structure

- Describe the purpose of each HTML page.
- Explain how users navigate between pages.

C. HTML and CSS Explanation

- Select at least three HTML elements used in your website and explain their purposes.
- Select at least three CSS properties and explain how they affect your website's appearance or layout.

D. JavaScript Explanation

For each interactive feature:

1. Explain what the feature does.
2. Include the relevant JavaScript code.
3. Explain how the code works.
4. Explain how JavaScript interacts with the HTML elements.

E. Reflection

- What was the most challenging part of this project?

- What did you learn?

- What would you improve if you had more time?

Your explanations must be written **in your own words**.

Do not simply copy definitions or explanations generated by AI.

4. Submission Requirements

Submit the following:

1. GitHub Repository URL

   - Include all HTML, CSS, JavaScript, and image files.

2. GitHub Pages URL

   - Your website must be publicly accessible and functional.

3. Code Explanation Report (PDF)

   - Include screenshots and explanations of your website.

Make sure all links, images, and interactive features work correctly before submission.

5. Grading Criteria

1. HTML Structure and Content 20%
2. CSS Design and Layout 20%
3. JavaScript Functionality 10%
4. Creativity and Usability 20%
5. Code Explanation Report 20%
6. GitHub Deployment 10%

6. Programming Restrictions

*****Only HTML, CSS, and JavaScript are allowed.****

The following are NOT allowed:

- Website builders (Wix, WordPress, etc.)

- Front-end frameworks (React, Vue, Angular, etc.)

- CSS frameworks (Bootstrap, Tailwind CSS, etc.)

- JavaScript libraries (jQuery, etc.)

- Pre-built website templates

All website structure, styling, and interactive functionality must be implemented using HTML, CSS, and vanilla JavaScript.

Academic Integrity and AI Usage

You may use AI tools as learning resources to help you understand programming concepts, debug errors, or explore possible solutions.

However, *you are responsible for understanding every part of the code you submit.*

Submitting AI-generated code without understanding how it works is not acceptable. Your project must reflect your own learning and programming abilities.

Individual Oral Examination and Offline Coding Test

After project submission, students whose work raises concerns about authorship or understanding may be invited to participate in:

1. An individual oral examination, during which you will be asked to explain your website and source code.

2. An offline coding test, during which you may be asked to write, modify, or debug code without access to the Internet or AI tools.

Students may be selected for verification if their submitted work contains unusually advanced techniques, differs significantly from their demonstrated programming abilities, or raises other reasonable concerns about authorship.

Being selected for verification does not automatically indicate academic misconduct.

However, students must be able to demonstrate that they understand and can explain the code they submitted.

Failure to demonstrate sufficient understanding may result in a grade adjustment or further academic integrity review.

- Be prepared to explain and modify your code.

Remember: A simple website that you fully understand is better than a complicated website that you cannot explain.